

CHAE EUN KIM

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Curriculum Vitae

Research Interests | *Biosensors; Bioelectronics; Point-of-care testing (POCT); Wearable sensing; Biosensor-integrated organ-on-chip systems*

Education

Seoul National University, Seoul, Republic of Korea

Feb 2026

- M.S. in Applied Bioengineering. Advisor: Prof. Kangwon Lee.
- Thesis: Calibration-Free Potassium Quantification via a Stoichiometric Enzyme Cascade on Laser-Induced Graphene (**Outstanding Thesis Award**)

Chung-Ang University, Seoul, Republic of Korea

Feb 2024

- B.S. in Biomedical Engineering.
- Thesis: Trends in Antigen-Antibody-Based Biosensor Technology for Pathogen Diagnosis.

Research Experience

Advanced Functional Biomaterials Laboratory, Seoul National University

Jul 2023 – *Present*

(Advisor: Prof. Kangwon Lee)

Researcher

Mar 2026 – Present

- Advanced the commercialization of a potassium-ion sensing platform through device integration, performance validation, and coordination with external testing institutions.
- Fabricated electrospun components for a MOTIE-funded 3D cell-based regenerative therapy project, supporting the development of materials and devices for cell preservation and thawing.

Graduate Researcher

Mar 2024 – Feb 2026

- Developed a calibration-free potassium sensor using a pyruvate kinase/pyruvate oxidase cascade on laser-induced graphene; led experimental design, optimization, and first-author manuscript preparation.
- Achieved a 2–8 mM linear range and inter-sensor RSD of 2.41–5.20%; validated performance in potassium-spiked human serum.
- Developed and evaluated disposable potassium-sensing strips with capillary-blood samples from maintenance-hemodialysis patients in collaboration with Severance Hospital; co-first author of the clinical study.
- Integrated carbon-coated δ -MnO₂ nanosheets into potassium-sensing strips with Ewha Womans University, contributing to a 1.4-fold faster response than activated-carbon contacts.

Undergraduate Research Intern

Jul 2023 – Feb 2024

- Worked on the development of potassium ion-selective electrodes, including electrochemical sensor fabrication, characterization, and optimization.

Translational Multiscale Biofluidic Laboratory, UNIST (Advisor: Prof. Joo Hun Kang)

Jul 2022

Undergraduate Research Intern (U-SURF program)

- Trained in magnetic blood cleansing, μ FISH-based bacterial identification, and vascularized microfluidic networks.

Department of Preventive Medicine, Seoul National University College of Medicine Jun 2021 – Aug 2021
(Advisor: Aesun Shin)
Undergraduate Research Intern.

- Reviewed literature on genetic variants associated with colorectal cancer susceptibility in Asian populations.

Publications

1. **Chaeun Kim**, Chaehyung Kim, Yujoon Kim, and Kangwon Lee*. Calibration-free potassium quantification via a stoichiometric enzyme cascade on laser-induced graphene. *Biosensors and Bioelectronics* 304, 118572 (2026). DOI: 10.1016/j.bios.2026.118572
2. Cheol Ho Park†, Yujoon Kim†, **Chaeun Kim**†, Chaehyung Kim†, Sowon Kim, Kangwon Lee, and Tae-Hyun Yoo*. Clinical evaluation of a disposable test strip for potassium level measurement in patients treated with maintenance hemodialysis. *Clinical Journal of the American Society of Nephrology* 21(3), 425–436 (2026). DOI: 10.2215/CJN.0000000934, †Equal contribution.
3. Jaeyeon Lee, **Chaeun Kim**, Chaehyung Kim, Yujoon Kim, Seojin Lee, Kangwon Lee*, and In Young Kim*. Supercapacitive carbon-coated δ -MnO₂ nanosheets as a solid-contact material for self-testing potassium-ion biosensors. *Small* 22(26), e13386 (2026). DOI: 10.1002/smll.202513386

Patents and Intellectual Property

1. **Strips for measuring glucose and potassium** Jan 2025
Yujoon Kim, **Chaeun Kim**, et al. Korean Patent No. 10-2757781 (granted). Patent record

Conference Presentations

1. **Chaeun Kim**, Chaehyung Kim, Yujoon Kim, Kangwon Lee*, “Polyvinyl Butyral (PVB)-Enhanced Reference Electrode for Stable Potassium Sensing in Point-of-Care (POC) Applications.” Korean BioChip Society – Poster presentation Nov 2024
2. **Chaeun Kim**, Chaehyung Kim, Kangwon Lee*, “Polyvinyl Butyral (PVB)-Enhanced Reference Electrode for Stable Potassium Sensing in Point-of-Care (POC) Applications.” NANO KOREA – Poster presentation Jul 2025
3. **Chaeun Kim**, Kangwon Lee*, “Long-Term Storage-Stable Potassium Sensing Enabled by a PVB-Enhanced Reference Electrode.” SNU BioDay – Poster presentation Sep 2025

Selected Projects

Simultaneous glucose and potassium test strip, Seoul R&D Program Sep 2023 – Aug 2025

- Project Manager, Seoul Business Agency (BT230224): led potassium-sensor experimental design, data analysis, and system integration; coordinated material–device research with Ewha Womans University.

Portable Potassium-Ion Measurement Device Development, THEDONEE Inc July 2023 – Present

- Led translational R&D for a portable potassium-ion measurement device and disposable test-strip platform.
- Coordinated sensor validation, measurement-circuit and firmware development, device housing design, and collaboration with external testing institutions.
- Contributed to prototype planning, performance verification, and clinical-use-oriented product development.

Honors and Scholarships

Outstanding Thesis Award , <i>Graduate School of Convergence Science and Technology, Seoul National University</i>	2025
Lecture & Research Scholarship , <i>Seoul National University</i>	Fall 2025
Best Poster Presentation Award , <i>Korean BioChip Society</i>	2024
Merit-based Scholarship , <i>Seoul National University</i>	Fall 2024
Audience Choice Award, ICC CORE-ATHON Startup Competition	Dec 2022
Scholarship Recipient , <i>Ulsan Institute for Talent and Lifelong Education</i>	2021
Academic Excellence Scholarship – Ranked 1st in Department (GPA: 4.5/4.5)	Fall 2019
Academic Excellence Scholarship	Spring 2019

Research & Technical Skills

Electrochemical Methods: Chronoamperometry (CA), Cyclic Voltammetry (CV), Potentiometry

Materials Characterization: Scanning Electron Microscopy (SEM), X-ray Diffraction (XRD), Raman Spectroscopy

Fabrication & Assay Development: Laser-Induced Graphene (LIG) Fabrication, O_2 Plasma Treatment, Enzyme-Cascade Optimization, Ion-Selective Electrodes (ISEs) Fabrication, Electrospinning

Optical Analysis: UV–Vis Spectrophotometry, Microplate-Based Absorbance Assays

Analytical & Clinical Validation: Sensor Calibration, Selectivity and Reproducibility Testing, Clinical Sample Analysis

Software: R, OriginPro, GraphPad Prism, ImageJ, AutoCAD